CHAPTER

Structural bioinformatics

6

6.1 Introduction

The study of biology is done with the help of computer science and programming in the discipline of bioinformatics, which is an example of an interdisciplinary field. The advancement of biological research results in the collection of enormous volumes of biological data, which is difficult to process if one does not have access to computational capacity. When it comes to manipulating biological components, bioinformatics has proven to be a very useful tool. A subfield of bioinformatics known as structural bioinformatics investigates the structure of biological macromolecules and micromolecules, including things like protein, DNA, and RNA, among others. A living object is composed of a significant number of macromolecules. Therefore, in order to analyze them, you need to understand how they are constructed. The study of structural bioinformatics reveals to us how these components are assembled, so it assists us in learning more about them. Bioinformatics is an essential interdisciplinary topic that helps with storing, processing, and analyzing data in a methodical and comprehensive fashion. This is because there is a lot of biological data that is being made public. The use of bioinformatics approaches has made it simpler to manipulate biological data such as DNA sequences and protein sequences, thanks to the contributions of statistics and algorithms to the field. When compared to sequential data, biological structures are often more fluid and unpredictable. Because of this, the field of structural biology presents a more significant challenge than the field of bioinformatics.

Another part of structural bioinformatics is predicting the structure of biological macromolecules. Bioinformatics tools and algorithms can be used to figure out the structure of a protein. Proteins are the body's essential structural and functional components. Proteins are used to make hormones, which control how metabolism works. Proteins are also used to make hair, muscle fibers, antibodies, and wool, among other things. Enzymes are the most important part of all biochemical reactions that happen inside the body. They also help our bodies fight off infections, turn food into energy, and help with the replication, transcription, and translation of DNA. In general, about 60% of a person's body is made up of water, and 17% is made up of proteins. To do their jobs, proteins need to be folded into a certain shape and structure. By breaking down how proteins fold, researchers can better understand what they do

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in the body and might be able to make protein supplements for people who don't get enough. Misfolded peptides can give us more information about how diseases spread. A lot of work is being done to come up with new ways to test how proteins look in three dimensions. Researchers often use X-ray crystallography and NMR spectroscopy to deduce the structure of proteins. But the experimental methods cost a lot of money and take a long time. A protein's structure can be predicted based on its sequence or a similar structure.

Structural bioinformatics usually consists of the following steps:

- Collecting of biological data, either high throughput sequencing data or imaging data.
- Building a computational model can be a structural simulation, optimization, or alignment.
- Interpreting the model results from structural biology perspectives.
- Providing insights for the next iteration of experimental design.

6.2 Viewing protein structures

Visualization drives a cycle of experimentation, reasoning, speculation, and validation in many areas of research, particularly structural biology and biophysics. The use of molecular visualization in particular is now common in a variety of situations, whether it is to illustrate scientific research articles or to gain understanding of original research data. These techniques have long attracted widespread interest and demand. Data transfer from fundamentally three-dimensional objects to a 2D framework, such as paper or standard computer displays, is a significant challenge. The computer graphics discipline has made significant contributions to computer science. Because the field of structural biology requires easily accessible, end-user focused software tools for effective dissemination, these contributions could only arrive gradually. The long-standing need for molecular graphics has given rise to macromolecular structure visualization tools, which are now available to scientists of all backgrounds.

Molecular vaporization has a wide range of applications, from extremely general needs to specialized ones. A first generic application, for example, is the evaluation of hypotheses and the representation of scientific papers, such as when correlating mutational data with structural representations. The next level of application could be a more detailed visual examination of macromolecular structures and their attributes, potentially in relation to the (spatial) distribution of charges, electrostatic properties, pockets, and surface complementarities. An even more specialized application could be the representation and analysis of data using theoretical chemistry, computational biology, and bioinformatics methods. One notable example that has advanced the field is the need to examine increasingly complex molecular dynamics simulations. This application invariably leads to data analytics, specifically visual analytics.

Several molecular visualization software programmes have been in use by the community for many years and provide powerful visualization features to a large user base. Popular molecular visualization softwares include Chimera (the most recent version is called ChimeraX) (Pettersen et al., 2004), JMol (and more recently JSmol) (Herráez, 2006), PyMol (Yuan et al., 2017), and VMD (Humphrey et al., 1996). These legacy packages appear to be a safe choice for visualization-based applications, with a solid codebase and an accessible API. Although not always intuitive for beginners, their utility has grown over time, and they are well suited to molecular visualization tasks. Each tool may have a variety of features and provide extensions, scripts, and tutorials to improve the user's experience. Table 6.1 provides a brief overview of this indispensable set of software tools, while Fig. 6.1 depicts a visual comparison of a typical workday and the user interfaces of the various programmes. The Chimera package, originally known as Midas in 1976, provides features for visual exploration and the study of molecular structures. It is possible to examine related descriptive data, particularly cryo-EM datasets with density maps, molecular dynamics trajectories, and other data. It is easy to make animations of the systems shown, and there are many different ways to show them visually.

Jmol was founded in 1999. It is a versatile platform that can be used as a standalone viewer as well as in a web environment. Jmol is often used to show structures in educational applications because it is flexible and easy to add to courseware or use as a graphical interface for looking into structural databases.

PyMol, as the name implies, was released near the end of 1999 and is based on the Python scripting language. This tool is especially popular among experimentalists due to the properties it provides that are useful for crystallographic and NMR-derived structures. It generates illustrations suitable for publication and includes useful molecular editing and atom selection tools. The term VMD, or Visual Molecular Dynamics, refers to the software's primary focus since its inception in 1993, which has been the graphical interpretation of molecular dynamics data. It efficiently maintains even large systems and provides a wide range of enhancements, particularly for sophisticated visual analysis, via plugins.

All of these programmes have the ability to automate and script operations for easy re-use and bulk deployment, which is a critical feature. Despite their long

Table 6.1 Summary of four commonly used protein visualization tools.

Features/ Tools	VMD	Pymol	Chimera	JMol/JsMol
Scripting	TCL	Python	Python	Javascript
OS	Windows/Mac/Linux	Windows/ Mac/Linux	Windows/Mac/ Linux	Web, Windows/ Marc/Linux
URL	https://www.ks.uiuc.edu/Research/vmd/	https://pymol.org/2/	https://www.cgl.ucsf.edu/chimera/	http://jmol.sourceforge.net/

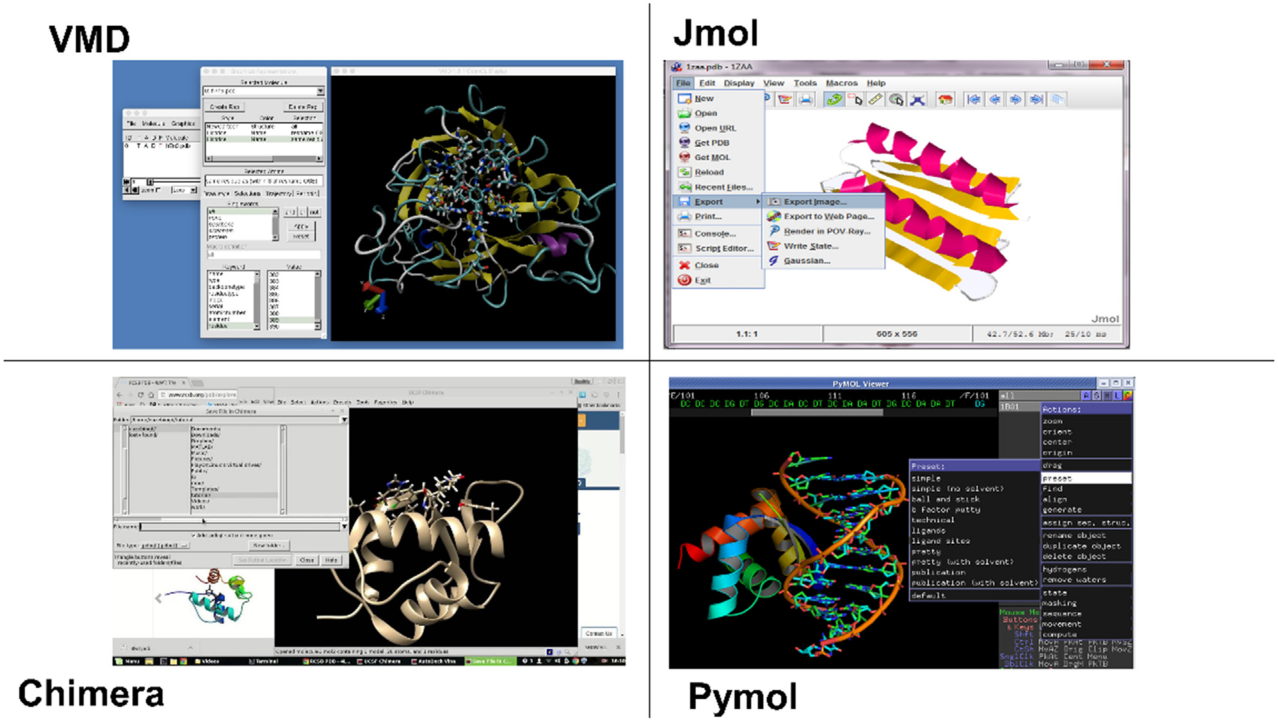

FIG. 6.1

Various protein visualization tool.

histories, all of the packages continue to add new features and adapt to changing hardware innovations, such as those in the graphics card market. Sometimes, as with ChimeraX, these additions necessitate a significant redesign of the underlying code.

6.3 Alignment of protein structures

The approach that is based on templates is the most trustworthy method for predicting the structure of macromolecules. At the moment, there are three distinct varieties of alignment-based strategies: those that are based on sequence, those that are based on profiles, and those that are based on structures. In most cases, the sensitivity of the profile-based alignment method is much higher than that of the sequence-based alignment systems. The structure-based alignment technique, on the other hand, is more sensitive than the profile-based alignment system. When it comes to matching, sequence-based tactics provide a higher level of specificity compared to profile-based strategies. In the same way that structure-based alignment methods are more particular, profile-based alignment tactics are as well.

When two proteins are identical, the sequence-based techniques will provide a better alignment than the other options. There are now a number of different techniques being developed, and the primary distinction between them is the operation of their alignment, gap penalty, and mutation score of amino-alkanoiate algorithms. Some alignment techniques, such as the Needleman-Wunsch algorithm for global alignment and the Smith-Waterman algorithm for local alignment, create the alignments via the use of dynamic programming. Other alignment strategies, such as the Smith-Waterman algorithm, generate the alignments locally. FASTA and BLAST are two examples of additional approaches that make use of alignment algorithms that are heuristically based. In order to determine the degree of similarity between two aligned residues, PAM and BLOSUM compares the aminoalkanoic acid substitution matrices that are extensively employed. Therefore, it is possible that two proteins with just a passing resemblance in their sequence would need to have the same structure in order to be considered homologous. The alignment standard is going to be strengthened with the help of the operational sequence profile. Multiple sequence alignment, also known as MSA, was used in the creation of the sequence profile by utilizing homologous sequences. Not only does it reveal the aminoalkanoic acid sequence, but it also gives information about other chemical processes. PSI-BLAST (Jones and Swindells, 2002) can be used to find close homologs of the target macromolecule in a large set of sequence data, such as the non-redundant NCBI data, so that an MSA can be built from these homologous elements, and then a sequence profile can be created from the multiple sequence alignment. PSI-BLAST can also be used to find close homologs of the target macromolecule in a large set of sequence data. There are several approaches to aligning the main sequence with a sequence profile, as well as numerous approaches to aligning two sequence profiles with one another. The process of matching a primary sequence to an HMM profile requires the use of two separate tools called HMMER and SAM square. The programmes "DIALIGN" (Morgenstern, 1999) and "FFAS" (Jaroszewski et al., 2011) are two further examples of sequences in profile alignment software. FORTE, HHpred, and Sculptor are examples of software programmes that are capable of user profile-to-profile alignment. Systems that move from one step to the next or that go from one step to a profile need to function less poorly than these ones do. In order to create a profile for a particular sequence of peptides, BLAST-PSI will be used. A profile-dependent alignment procedure likewise makes use of the sequence illustration profile as a standard measurement. PSI-BLAST reveals that a sequence-based profile is either a position-specific scoring matrix (PSSM) or a position-specific frequency matrix (PSFM). These matrices are heavily utilized in a variety of tools, such as those used for discovering similarities, determining folds, and predicting the structural properties of peptides. The length of the peptide sequence is denoted by N, and each position-specific scoring matrix (PSSM) and position-specific frequency matrix (PSFM) has 20N dimensions. When a PSFM is performed, each column maintains a record of the frequency with which 20 amino acids appear at a certain location in the sequence. Therefore, each column in a PSSM has the potential to transform into one of 20 unique amino acids in the same place. A reliable sequence-based profile will incorporate the maximum amount of information that can be included within the MSA. A sequence profile's quality is dependent not only on the samples it contains but also on factors such as the number of iterations of PSI-BLAST and the E-value cutoff that is used for determining whether or not two peptides are identical. When converting the frequency of aminoalkanoic acid to mutation potential, it further requires a method that may include pseudocounts of aminoalkanoic acid. Profile Hidden Andrei Markov A different approach is used when modeling a multiple sequence alignment of homologous peptide sequences. The Hidden Andrei Markov Model is superior to the PSFM and PSSM because it explicitly models gaps in addition to taking into account correlations between neighboring residues. PSFM and PSSM only take into account correlations between adjacent residues. When it comes to aligning macromolecules and locating distant similarities, this indicates that profile HMM is, on average, more sensitive than PSFM/PSFM. A profile HMM might be in one of three states: match, insert, or delete. This is the most significant aspect. The "match" status of an MSA column indicates the likelihood that residues will be permitted to remain in the column once the analysis has been completed. and the next.

There are numerous tools used for the alignment of protein structure like-MADOKA (Deng et al., 2019), iPBA (Gelly et al., 2011), protein tertiary structure, pairwise sequence alignment, multiple sequence alignment, etc. (Figs. 6.2 and 6.3).

6.4 Structural prediction

Through the use of sequence similarity searches, multiple sequence alignments, identifying and describing domains, predicting secondary structure, predicting

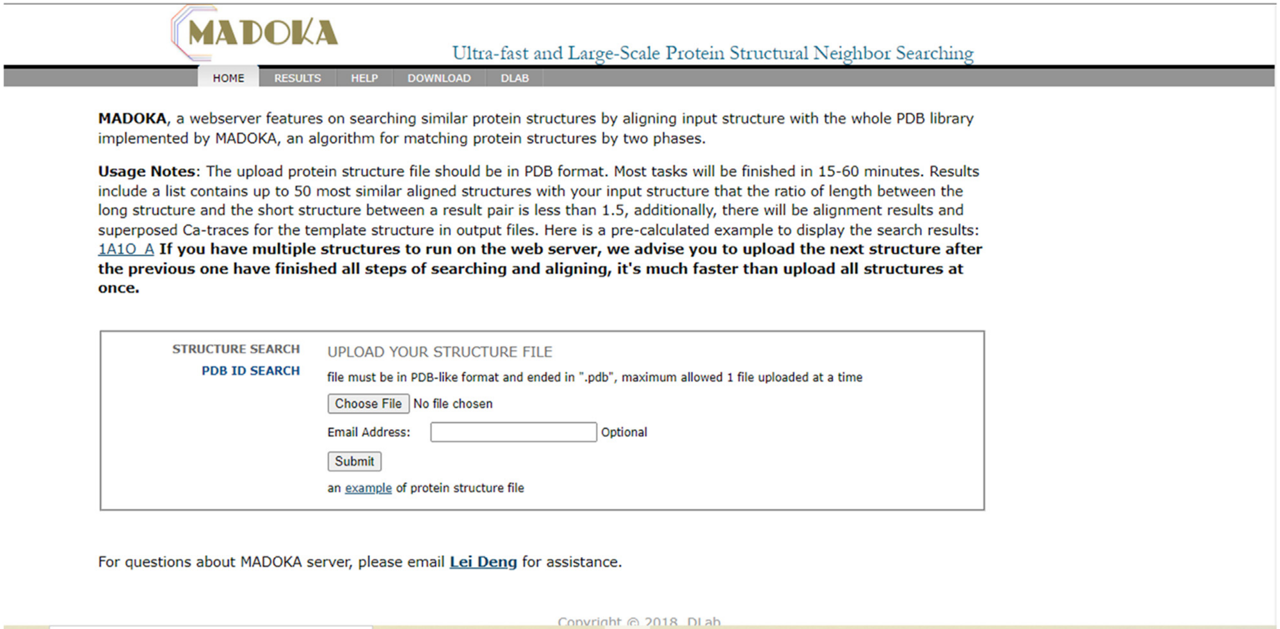

FIG. 6.2

Image of MADOKA webpage.

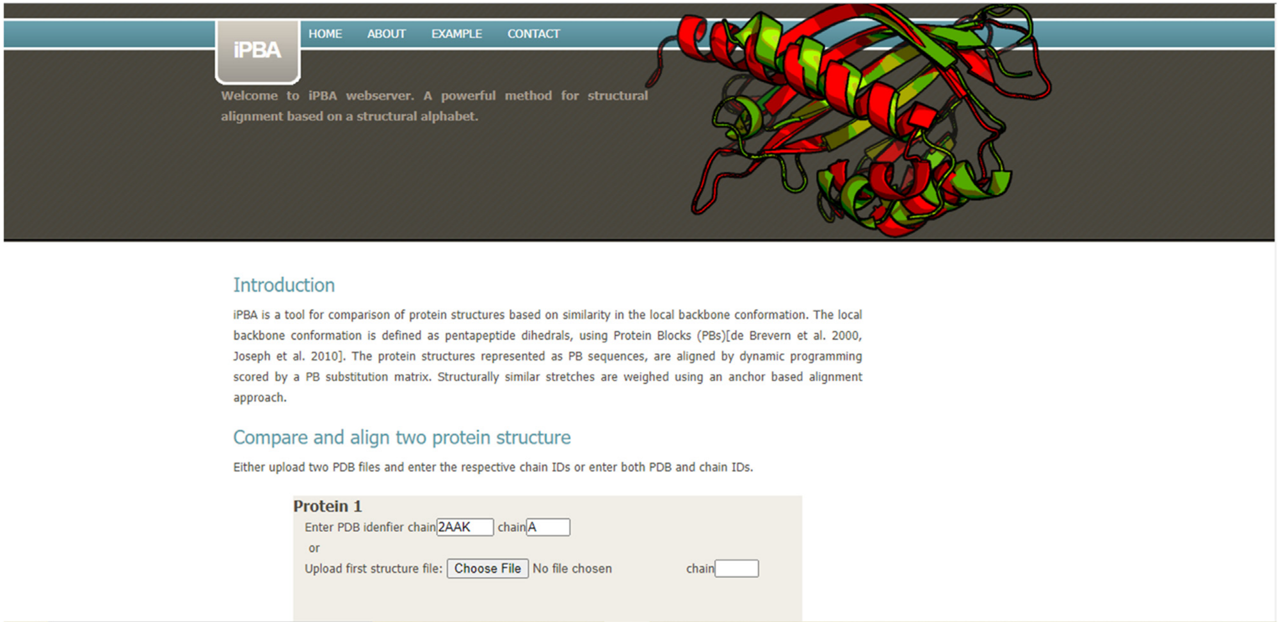

FIG. 6.3

Image of iPBA webpage.

solvent accessibility, automatically recognizing protein folds, building three-dimensional models down to the atomic level, and validating models, bioinformatics can be used to predict structures. Validating models is another application of bioinformatics. All of these approaches are not being used in every investigation into the composition of a protein's three-dimensional structure. Finding a reliable structural target from which to infer information about a protein's three-dimensional shape in response to a query sequence is an essential aspect of the process of protein structure prediction. There are three distinct types of schemes, each of which is determined by how it is carried out. The first step is to implement methods that are standard and well-known in the industry. In the event that a structural template cannot be found, the second effort has to make use of procedures that are more difficult. If a target fold cannot be defined in a reliable manner because many studies of nontrivial data have yielded varying findings, the design falls into the third category, and it will be very complicated, if not impossible, to complete with any degree of dependability.

Experiments that scientists do in order to determine the sequence and structure of proteins have seen a lot of innovation and improvement in the course of the previous few decades. Since the 1990s, the amount of protein data that has been uploaded to UniProt2 and the Protein Data Bank (PDB) has increased at a pace that is virtually exponential. When compared to obtaining the structures of proteins, obtaining their sequences is a much simpler task. The rapid accumulation of protein sequence data may be attributed to the advent of more sophisticated technologies for DNA sequencing. The UniProt/TrEMBL database now contains more than 85 million different protein sequences. X-ray crystallography and nuclear magnetic resonance (NMR) spectroscopy are the two primary experimental approaches that are used to find out how proteins are put together. Both of these methods focus on the structure of proteins. But both methods require a lot of time and work, and each has its own technical limitations when it comes to studying the proteins they want to study (Table 6.2).

6.4.1 Use of sequence patterns for protein structure prediction

The sequences of a genome include a wealth of information on the function of proteins and the ways in which they have evolved through time. This knowledge may be put to use to discover evolutionary connections between protein residues and to address the age-old challenge of determining the three-dimensional structure of a protein based on its amino acid sequence. Recent work on this issue has resulted in a significant amount of improvement, which can be attributed to the rapidly increasing number of sequences that are now accessible as well as the use of global statistical approaches. A better understanding of covariation, in addition to the three-dimensional structure, may help in the search for functional residues that are involved in the formation of protein complexes, the binding of ligands, and changes in conformation. The computer prediction of protein structures, which has been an issue in molecular biology for more than 40 years, may be able to fill this gap if it can be done with adequate precision. By comparing the amino acid sequence of interest

Table 6.2 Tools for protein prediction.

Name	Used for	Description
IntFOLD (Mcguffin et al., 2019)	Used for prediction of tertiary structure—3D modeling, domain prediction	The 3D structure and function may be predicted from amino acid sequences using an automated and integrated workflow.
RaptorX (Wang et al., 2016)	Is used for detection of remote homology	Progressed in predicting 3D structure whose protein sequence is without close homologous in PDB
Biskit (Grünberg et al., 2007)	3D modeling, template identification, and alignment	Predicting protein structure using homology modeling
ESyPred3D (Lambert et al., 2002)	Protein design and energy calculations	Using neural networks, the technique provides better alignment. Incorporating, weighting, and filtering various alignment processes provides alignments.
Rosetta (Baek et al., 2021)	Short fragments of known proteins are assembled by a Monte Carlo strategy to yield native-like protein conformation	De novo protein structure prediction
AlphaFold (Jumper et al., 2021)	Protein structure prediction	An artificial intelligence program developed by DeepMind, a subsidiary of alphabet, which performs predictions of protein structure.

to the sequence of another protein whose three-dimensional structure is known, several useful and fairly accurate three-dimensional models have been generated from amino acid sequences. These models may be used to better understand how proteins work. Building a model using a template, is often known as homology modeling. It has been proven difficult to produce good de novo predictions from the sequence when there is no known structure in a protein family. To fold proteins from scratch, some of the best, most current, and most cutting-edge methods, such as Rosetta, involve scanning for fragments with similar sequences in databases of three-dimensional structures and then utilizing empirical intermolecular force fields to put those fragments together. These kinds of approaches have been shown to be effective for proteins containing less than 90 amino acids, but in order to analyze bigger proteins, they need to be paired with the results with iterative tests of predicting structure of small residue patches. Other methods make an effort to generate 3D structure through predicting residue contacts by combining three-dimensional information with machine-learning techniques such as support vector machines, random forests, and neural networks. Despite these efforts, contact prediction accuracy remained "still quite low," and substantial improvements to models were only achieved for some small proteins. The issue of de novo structure prediction does not scale, meaning that the conformational search space grows exponentially as the size of the protein does. This presents a basic computational hurdle, even for approaches that are based on fragments of the protein. In this way, the main problem of predicting de novo structures in three dimensions has not been solved satisfactorily.

6.4.2 Prediction of protein secondary structure from the amino acid sequence

The local structure that is formed by a protein's polypeptide backbone is referred to as the protein's secondary structure. The -helix (H), the -strand (E), and the coil (C) area are the three different kinds of secondary structures. While states H and E both exhibit regular behavior, the C state does is disordered. The Dictionary of Secondary Structure of Proteins (DSSP) (Kabsch and Sander, 1983) was the product of Sander's efforts to develop a system for the assignment of secondary structure. The patterns of hydrogen bonds are used by this approach to automatically categorize the secondary structure into eight different states, which are labeled H, E, B, T, S, L, and G. Most of the time, these eight states may be further simplified into only three categories: helix, sheet, and coil. Usually, the helix is referred to by the letters G, H, and I; the sheet is referred to by the letters B and E, and all other states are referred to as a coil. The challenge of predicting the secondary structure of a protein appears as follows: given the sequence of amino acids that make up the protein, determine whether or not each amino acid is located in the α-helix (H), β-strand (E), or coil area (C). The prediction of the protein's secondary structure is often evaluated based on its Q3 accuracy. This metric, which evaluates the proportion of residues for three-state secondary structures, determines whether or not the structures have been successfully predicted.

In 1951, Pauling and Corey made the hypothesis that the backbones of protein polypeptides might assume helical or sheet-like forms. This was in the days before the first structure of a protein was discovered. A great number of statistical and machine learning approaches have been developed in order to determine the secondary structure. One of the first approaches to predicting the secondary structure of a protein combined statistical analysis with heuristic analysis. In order to overcome the issue of correctly forecasting the secondary structure, the GOR technique makes use of information theory. The position-specific scoring matrix (PSSM), which comes from PSI-BLAST, shows how proteins have changed over time and what has caused the biggest changes in how secondary structures of proteins are predicted. Protein composition is the consideration of a shape along with 3 protein components from an amino acid series—the prediction of the second and higher structure from the primary. The prediction of the structure is unique to the ever-changing hassle of protein formation. Guessing protein composition is one of the most vital dreams pursued by way of laptop biology; and is critical in remedy (for instance, in the manufacture of medicine) and biotechnology.

6.4.3 Chou Fasman method

The Chou-Fasman method (Chou and Fasman, 1978) is a computational method for predicting the secondary structure of proteins from their primary sequence. Protein secondary structure refers to the local, regular conformations adopted by the polypeptide chain, which are formed by the interactions between the peptide bonds and the side chains of the amino acids. The main types of secondary structure are the alpha helix, beta sheet, and random coil. The Chou-Fasman method is based on the idea that the amino acid sequence of a protein can influence its secondary structure. Specifically, the method uses statistical analysis to identify patterns in the distribution of amino acids in known protein structures and then uses these patterns to predict the secondary structure of a protein from its primary sequence.

To perform the prediction, the protein sequence is first divided into overlapping windows, and the amino acid composition of each window is analyzed. The method then assigns a score to each amino acid based on its propensity to be found in different types of secondary structure. For example, amino acids with hydrophobic side chains, such as leucine and valine, tend to be found in alpha helices, while amino acids with polar side chains, such as serine and threonine, tend to be found in beta sheets. The method then uses these scores to calculate the likelihood that a particular amino acid will be found in a particular type of secondary structure. This likelihood is expressed as a probability, and the secondary structure with the highest probability is predicted for each amino acid. One of the advantages of the Chou-Fasman method is that it can be applied to proteins of any size and is relatively easy to implement. However, the method has some limitations. First, it relies on statistical analysis and may not always be accurate. Second, it can only predict the secondary structure of a protein, not its tertiary structure, which refers to the three-dimensional shape of the protein as a whole. Finally, the method does not take into account the effects of the protein's environment, such as pH and temperature, which can also influence its secondary structure. Despite these limitations, the Chou-Fasman method remains a widely used tool in protein structure prediction and has contributed to our understanding of the relationship between a protein's primary sequence and its secondary structure. It has also been used in combination with other methods, such as hydrogen bonding analysis and structural alignments, to improve the accuracy of protein structure prediction.

6.4.4 GOR method

The GOR (Gödel, Or, and Roth) (Garnier et al., 1978) method is a widely used method for predicting the secondary structure of proteins from their amino acid sequence. It is based on the idea that the local sequence of a protein is related to its local conformation, and that the conformation of a protein can be predicted by looking at the sequence of amino acids around it. The GOR method uses a statistical approach to predict the secondary structure of a protein based on the sequence of its amino acids. It does this by building a statistical model based on a set of known protein structures, and then using this model to make predictions about the secondary structure of a given protein. The GOR method begins by dividing the protein into overlapping segments, called windows, and then looking at the sequence of amino acids within each window. It then compares the sequence of each window to the sequences of windows in known protein structures, and uses this information to predict the secondary structure of the protein. The GOR method has a number of advantages. One of the main advantages is that it is relatively fast, as it only requires a single pass through the protein sequence. Additionally, the GOR method is relatively accurate, with an average prediction accuracy of around 70%.

There are also a number of limitations to the GOR method. One of the main limitations is that it only predicts the secondary structure of a protein, and does not provide any information about the tertiary structure or overall folding of the protein. Additionally, the GOR method is based on statistical models, and is therefore subject to the limitations of such models. Finally, the GOR method may not work well for proteins with unusual or atypical sequences, as it is based on a set of known protein structures and may not be able to accurately predict the structure of proteins that do not fit this set. Overall, the GOR method is a useful tool for predicting the secondary structure of proteins, and has played a significant role in the study of protein structure and function. It is particularly useful for predicting the secondary structure of proteins in the early stages of protein structure prediction, and can provide valuable information about the local conformation of a protein. However, it is important to be aware of its limitations, and to use it in conjunction with other methods in order to obtain a more complete picture of protein structure (Fig. 6.4).

6.4.5 Prediction of three-dimensional protein structure

Predicting the three-dimensional (3D) structure of a protein from its amino acid sequence is a central problem in computational biology. The 3D structure of a protein plays a critical role in its function, and determining the structure of a protein can provide insights into how it performs its function, how it may be modified or regulated, and how it may interact with other molecules. There are several experimental techniques that can be used to determine the 3D structure of a protein, such as X-ray crystallography and nuclear magnetic resonance (NMR) spectroscopy. However, these techniques are time-consuming and costly, and may not be feasible for all proteins. In addition, many proteins do not crystallize well or are too large to be studied by NMR, making it difficult to determine their 3D structure using these techniques.

Computational methods offer a faster and potentially more cost-effective alternative for predicting the 3D structure of a protein. These methods can be broadly classified into two categories: homology modeling and de novo prediction. Homology modeling involves using the 3D structure of a related protein, or "template," as a starting point to predict the 3D structure of a protein of interest. This approach relies on the assumption that proteins with similar amino acid sequences will have similar 3D structures. The protein of interest is first aligned with the template, and the resulting sequence alignment is used to build a model of the protein's 3D structure.

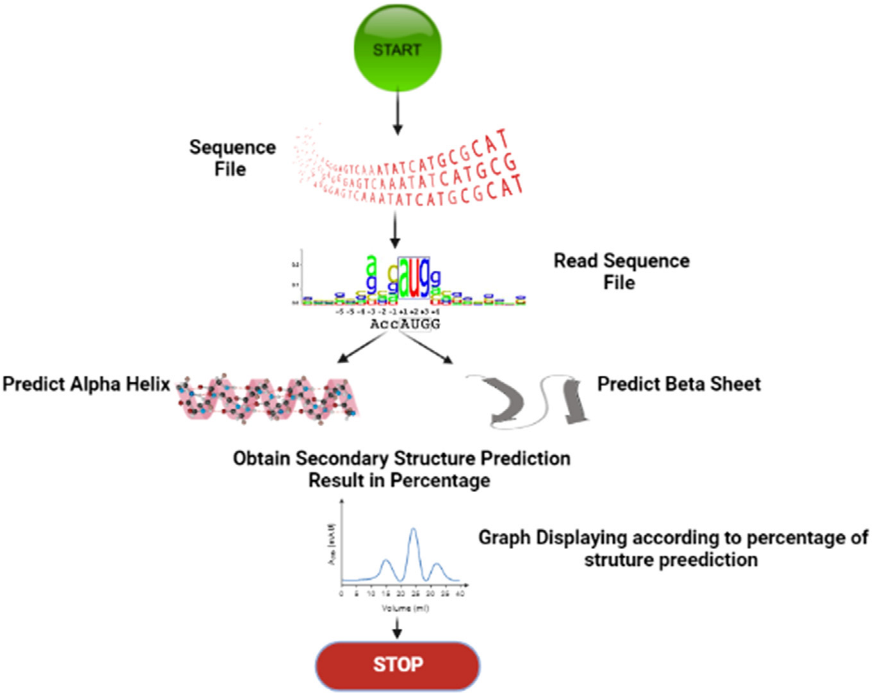

FIG. 6.4

GOR method.

Homology modeling can be an effective approach for predicting the 3D structure of a protein if a suitable template is available and the sequence identity between the protein of interest and the template is high.

De novo prediction, on the other hand, involves predicting the 3D structure of a protein from scratch without using a template. This approach relies on the physical and chemical properties of the amino acids and the interactions between them. De novo prediction methods can be divided into two main categories: physics-based methods and knowledge-based methods. Physics-based methods use physical principles, such as energy minimization, to predict the 3D structure of a protein. These methods are based on the idea that the protein will adopt a conformation that minimizes its energy. Knowledge-based methods, on the other hand, use statistical information about the 3D structures of known proteins to predict the 3D structure of a protein. These methods rely on the assumption that proteins with similar amino acid sequences will have similar 3D structures. Threading, also known as fold recognition, involves searching for the best fit of the target protein's amino acid sequence onto a library of known 3D protein structures. This approach is relatively fast and can be accurate, but it is limited by the size and quality of the library of known protein structures.

There are a number of different algorithms and software tools available for predicting the 3D structure of a protein using computational methods. Some of the most commonly used methods include ROSETTA, Modeler, and I-TASSER. These tools are widely used by researchers in academia and industry to predict the 3D structure of proteins and to understand their function.

Despite advances in computational methods, predicting the 3D structure of a protein remains a challenging problem. Accurate prediction of protein structure can provide valuable insights into the function of a protein and can be useful in drug design and other applications. However, current methods are often limited in their accuracy, and further research is needed to improve the prediction of protein structure. While these methods are not perfect and cannot always accurately predict the 3D structure of a protein, they offer a valuable alternative to experimental techniques and have the potential to significantly accelerate our understanding of protein structure and function (Fig. 6.5).

6.4.6 Evaluating the success of structure predictions

Evaluating the success of computational protein structure predictions is an important aspect of the protein structure prediction field, as it allows researchers to determine the accuracy and reliability of different prediction methods. There are several different ways to evaluate the success of protein structure predictions, and the most appropriate method will depend on the specific goals of the prediction and the type of data available. One common method for evaluating the success of protein structure predictions is to compare the predicted structure to the experimentally determined structure, using a metric known as the root mean square deviation (RMSD). The RMSD measures the average distance between the atoms in the predicted and experimental structures, and a lower RMSD indicates a more accurate

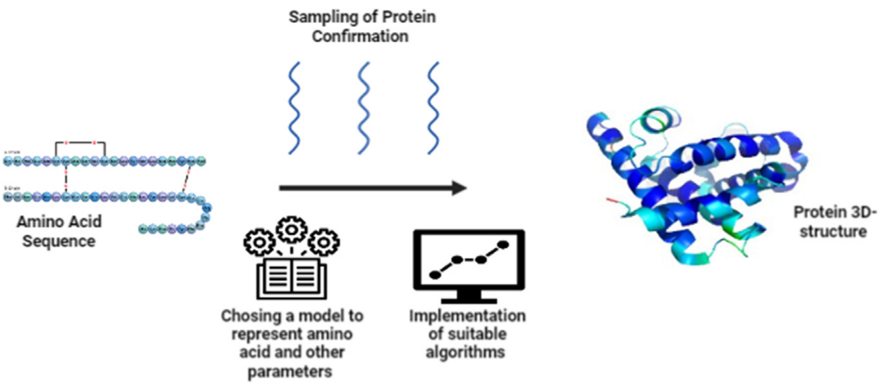

FIG. 6.5

3D protein structure prediction.

prediction. The RMSD can be calculated for all atoms in the protein, or for a subset of atoms such as the backbone atoms or side chain atoms.

Another metric that is often used to evaluate protein structure predictions is the precision and recall of the prediction. Precision measures the fraction of predicted residues that are correctly predicted, while recall measures the fraction of correctly predicted residues out of all residues in the protein. A high precision and recall indicates a more accurate prediction. Another way to evaluate protein structure predictions is to use a benchmarking dataset, which consists of a set of proteins for which the experimental structure is known. The prediction method is applied to each protein in the dataset, and the accuracy of the prediction is compared to the experimental structure. This allows researchers to compare the performance of different prediction methods on a standardized dataset. In addition to these quantitative measures, it is also important to consider the biological relevance of the predicted structure. A prediction that is highly accurate according to the RMSD or precision and recall metrics may not necessarily be biologically relevant if it does not correctly capture important features of the protein such as its active site or binding site. Overall, evaluating the success of computational protein structure predictions is a complex task that involves considering a range of different metrics and factors. By carefully considering the accuracy, precision, and biological relevance of the prediction, researchers can determine the effectiveness of different prediction methods and identify areas for improvement.

EVA (Eyrich et al., 2001) is a web server that evaluates the effectiveness of automated approaches for predicting the three-dimensional structure of proteins. The assessment is set to be automatically updated once per week to ensure that it remains current despite the ever-increasing number of prediction servers and the ever-shifting methods for making predictions. Every day, the servers connect to the Protein Data Bank (PDB) to get the sequences of newly discovered protein structures. Their predictions are then put together.

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